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Palaeomagnetism of the Salla Diabase Dyke, northeastern Finland, and its implication for the Baltica–Laurentia entity during the Mesoproterozoic

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Abstract: New palaeomagnetic and rock magnetic results are presented for the 1122 ± 7 Ma Salla Diabase Dyke in NE Finland. A positive baked-contact test proves that the dyke has a primary natural remanent magnetization carried by magnetite. The characteristic remanent magnetization direction ($D = 42.2^\circ$, $I = 73.9^\circ$, $k = 75.7^\circ$, $\alpha_{95} = 4.8^\circ$) of 13 sites along the large single dyke provides a virtual geomagnetic pole (VGP) position of Plat = 71°N, Plon = 113°E ($A_{95} = 8.1^\circ$). Although secular variation may not have been fully averaged out, the new VGP provides an important result to define the late Mesoproterozoic position of Baltica. The VGP is not close to any known Proterozoic palaeopoles of Baltica, and therefore the pre-Sveconorwegian apparent polar wander path (APWP) of Baltica must be modified. The pre-Sveconorwegian (*c*. 1.3–1.0 Ga) APW swathes of Baltica, Laurentia (including the Logan Loop) and Kalahari cratons show similar shape, but new well-dated palaeomagnetic poles for *c*. 1.25–1.12 Ga interval from these continents are required to test the similarity. The Salla dyke VGP provides hints that the Mesoproterozoic Baltica–Laurentia connection in the Hudsonland supercontinent assembly lasted until 1.12 Ga.

Mafic dyke swarms are useful rocks for palaeomagnetism since they generally carry a stable and primary remanent magnetization and they can be accurately dated with high precision using the U–Pb method (Buchan & Halls 1990). Coupled with studies of dyke orientations and information of the anisotropy of magnetic susceptibility, the mafic dyke swarms yield valuable constraints for supercontinent research as well as for identification of large igneous provinces (Ernst & Buchan 1993). The WNW–ESE striking *c*. 150 km long and 60–100 m-wide Salla dyke of NE Finland (Fig. 1) represents a key target for palaeomagnetic study for several reasons. First, it has been precisely dated at 1122 ± 7 Ma (U–Pb, zircon; Lauerma 1995). Second, its pole position is important in refining the pre-Sveconorwegian plate movement of Baltica during 1.27–1.0 Ga, which then can be compared with the Laurentian and with the Kalahari drift histories at the same time. Moreover, the age of 1122 Ma is important since it generally marks the timing of the amalgamation of the Rodinia supercontinent assembly after the breakup of Hudsonland (also known as Columbia or Nuna). Finally, with this new pole for Baltica we are able to study the proposed long-lasting unity of Baltica and Laurentia which formed the core of the Mesoproterozoic supercontinent. For example, Pesonen *et al.* (2003) have suggested that Laurentia and Baltica were connected for *c*. 600 Ma from 1.83–1.27 Ga with a separation which took place at *c*. 1.2 Ga. Several other authors have suggested a variety of configurations and lifecycles of this supercontinent (Karlström *et al.* 2001; Meert 2002; Zhao *et al.* 2004, and refs therein) which somewhat differs from the next supercontinent Rodinia that existed *c*. 1000–750 Ma ago. The new palaeomagnetic data from the Salla Diabase Dyke of Baltica allow us to study the timing of tectonic transition between supercontinents Hudsonland and Rodinia.

Geology and sampling

The Salla Dyke is located in the northeastern part of the Fennoscandian Shield (Fig. 1). There, Palaeoproterozoic supracrustal rocks cover the Archaean basement forming the Lapland Greenstone Belt (LGB), which extends from northern Norway through central Finnish Lapland to western Russia. The Finnish part of the LGB records a sedimentary and volcanic evolution spanning several hundreds of

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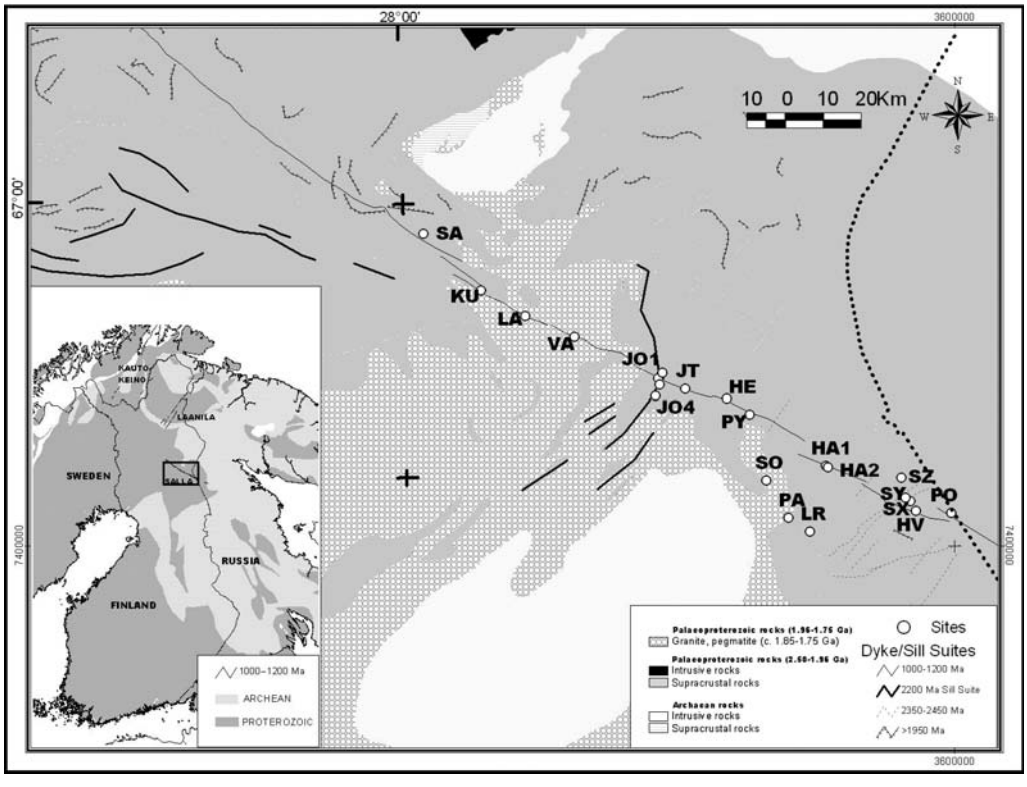


Fig. 1. Simplified geological map of the sampling area. The insert shows the study area (rectangle). Sampling sites are marked with letters. Dashed line marks the border between Finland and Russia.

millions of years (2.5–1.8 Ga) (Hanski & Huhma 2005). The WNW–ESE-striking Salla schist belt forms the southeastern extension of the Palaeoproterozoic LGB. This *c*. 100 km-long and 40 km-wide belt is divided by the Finnish–Russian border in a north–south direction (Manninen & Huhma 2001).

Metasedimentary rocks of the LGB in the Salla area are penetrated by a 2.2 Ga differentiated gabbro sill (Lauerma 1995; Hanski *et al.* 2001; Fig. 1). The upper part of the sill is composed of highly metamorphosed pyroxenite, which gradually changes to more gabbroic rocks that are the dominant rock type of the sill. At site Jokinenä (JO1-3, Fig. 1) the younger 1.12 Ga diabase dyke crosscuts the gabbro sill (Lauerma 1995). The contact of the younger dyke with the sill is visible at the northern side of the dyke.

The Salla Diabase Dyke is the youngest rock formation in the area, cutting the volcanic, metasedimentary and granitoid rocks (Lauerma 1995). The dyke forms an *en echelon* structure, and according to the aeromagnetic map it can be followed from the Russian side through Salla at least as far as central north Finland (Fig. 2), thus having a length of *c*. 150 km. The dyke is nearly vertical, it trends in a WNW–ESE direction, and its width varies typically from 60–100 m (locally up to 170 m). Some apophyses with widths of a few centimetres to decimetres have been found at site Jokinenä (JO1-3 in Fig. 1; Lauerma 1995). During the field campaign of 2006 we found a new *c*. 60 cm-wide apophysis from site Haltiavaara (HA2, Fig. 1).

The Salla Dyke has been described by Väänänen (1965), Lauerma (1995) and Manninen & Huhma (2001). The compositional structure of the dyke is symmetrical, and contacts with the host rocks (e.g. metavolcanic rocks and gabbro) are sharp. The fine-grained chilled margins have olivine basaltic composition that gradually changes to medium-grained layered pyroxene diabase. The main part of the dyke is glomeroporphyritic with phenocrysts of pyroxene and plagioclase, whereas the central part is reddish quartz diabase. Optical-microscopic analyses of the dyke indicate that flow differentiation is the main process causing the layered structure (Lauerma 1995).

Sampling of the Salla Dyke for palaeomagnetic studies began as early as in 1978. Since then samples have been taken during field campaigns

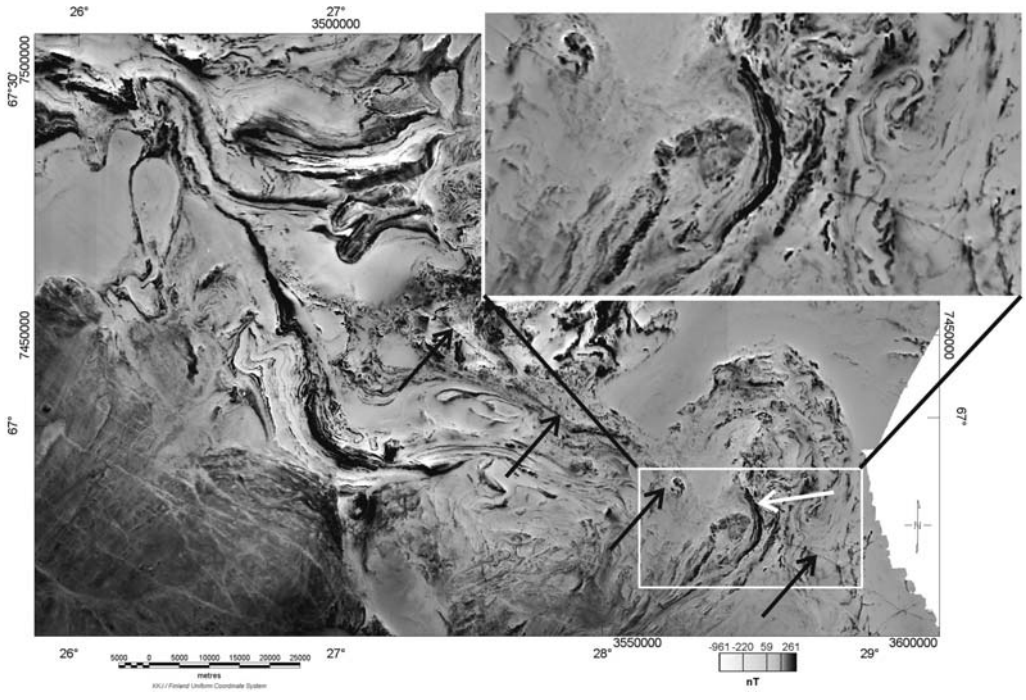


Fig. 2. Aeromagnetic total intensity map of Salla area, northern Finland. Black (white) arrows indicate the Salla Diabase Dyke (2.2 Ga gabbro sill).

in 1981, 1984, 1988, 1989, 2001, 2006 and 2007. Altogether 194 diabase cores and hand samples were collected from 15 sites. Orientation of the samples was made using either sun or magnetic compasses. Sampling was restricted due to the fact that there is poor exposure except on hills. Many of the outcrops are broken by frost action and turned to block fields. Sampling sites are shown in the geological map (Fig. 1).

In order to carry out baked-contact tests for the dyke, samples were taken also from the host rocks. Fifty-one metavolcanic samples (both baked and unbaked) from nine sites, twenty-eight gabbro samples (both baked and unbaked) from three sites, and six unbaked granite samples from one site were investigated. The 2.2 Ga gabbro sill at site Jokinenä 4 was sampled along a *c*. 200 m-long railway cutting (site JO4, Fig. 1), where the upper part of a 2.2 Ga layered sill can be seen, and at site Jokinenä 3 where the Salla Diabase Dyke cuts cross it (site JO1-3, Fig. 1).

Methods

Petrophysical and palaeomagnetic measurements have been carried out both at the Solid Earth Geophysics Laboratory of the University of Helsinki and at the Palaeomagnetic Laboratory of the Geological Survey of Finland.

Petrophysical properties, such as natural remanent magnetization (NRM), magnetic susceptibility and density, were measured for each specimen before palaeomagnetic and other rock magnetic studies. Remanence measurements were made using a 2G-Enterprise SQUID magnetometer and an Agico spinner magnetometer.

Palaeomagnetic studies included alternating-field (to peak fields of 160 or 100 mT) and thermal (up to temperatures of 680 °C) demagnetization of the specimens. Alternating field demagnetization was carried out using the 2G SQUID magnetometer static-coil system. Some samples were demagnetized using the Agico LDA-3 demagnetizer. Thermal cleanings were carried out with the single-axis Schonstedt TSD-1 furnace, the Magnetic Measurement thermal demagnetizer and the ASC thermal furnace of the palaeomagnetism laboratory of the Otago University in New Zealand. Measurement of magnetization was carried out using a 2G SQUID magnetometer, both in Finland and in New Zealand. During the thermal cleaning, the bulk magnetic susceptibility was monitored after each step in order to detect mineralogical changes

due to heating. Specimens from 114 diabase samples and from 63 host rock samples (9 andesites, 21 gabbros, 4 granites, 4 amphibolites and 15 metavolcanic rocks) were thermally cleaned. Remanence components were identified using principal component analysis (Kirschvink 1980; Leino 1991). Statistical means were calculated according to Fisher (1953). The palaeopoles were plotted with GMAP program (Torsvik & Smethurst 1999).

In order to identify magnetic carriers, thermomagnetic analyses (susceptibility v. temperature) were carried out using an Agico CS2-KLY3 kappabridge. To avoid oxidation during heating, the specimens were heated in argon gas, from room temperature to 700 °C and back to room temperature. Curie points were determined from the maxima in the second derivatives of the thermomagnetic curves (Tauxe 2002; Petrovsky & Kapicka 2006). For determination of domain states of the magnetic carriers, hysteresis properties were measured using a Princeton Measurement Corporation MicroMag^{TM}3900 model Vibrating Sample Magnetometer (VSM). Scanning Electron Microscope (SEM; Jeol JSM-5900LV) analyses were performed in order to verify the nature of the magnetic carriers. Rock magnetic properties of the ferromagnetic minerals were studied using the Lowrie test (Lowrie 1990), which distinguishes the test utilizing the different coercivities and thermal unblocking temperature characteristics of the most common ferromagnetic minerals. Isothermal remanent magnetization was given at 1.2, 0.4 and 0.12 T fields along the z, y and x axis, respectively. Samples were then thermally demagnetized up to 680 °C. After each step, magnetic susceptibility and remanence were measured.

The anisotropy of magnetic susceptibility was measured for a selection of diabase specimens using the Agico KLY-3 kappabridge in order to define the possible deformation of the dyke (Hrouda 1982) and to characterize the magma flow pattern (Rochette *et al.* 1992).

Stability of magnetization was studied using the simplified version of the viscosity test, which defines the changes in magnetization due to viscous processes (Dunlop & Buchan 1977). It consists of measuring the magnetization of the specimens before (initial NRM, NRM_0) and after (stable NRM, NRM_{st}), storing them in a zero-field environment for three weeks. The viscosity coefficient S_v (%) can be determined using the formula: $S_v = [(NRM_0 - NRM_{st})/NRM_0] \times 100$.

Results

Petrophysical properties

The Salla Diabases generally yield stable palaeomagnetic results, but they are only moderately strongly magnetized. Mean results of density, magnetic susceptibility, remanence intensity (NRM) and Koenigsbergers ratios (Q-values) for the diabase and host rocks are shown in Table 1. Figure 3 shows the NRM v. susceptibility with isopleths of Q.

There is little variation of bulk petrophysical properties between different diabase sites. Q-values are around 1–1.5 (Fig. 3), suggesting that the remanence of the samples is carried by moderate to large multidomain (MD) grains. The remanence of the larger grains is less stable and can more likely acquire viscous remagnetization (Dunlop 1983). However, the intensity of the natural remanent magnetization of the samples is moderately high (average 695 mA/m; Fig. 3; Table 1) and indicates that the original magnetization may be preserved.

Petrophysical properties of the diabases and host rocks differ. Unbaked metavolcanic rock samples have two-orders-of-magnitude lower susceptibility and remanence intensity values than the diabase samples (Fig. 3). Q-values of such rocks are low, indicating that they do not contain stable magnetization, as was also observed during the palaeomagnetic study. Baked metavolcanic rock samples show lower density and two-orders-of-magnitude higher susceptibility and remanence values than unbaked metavolcanic samples, indicating that baking has clearly affected the mineralogy and composition of such rocks. Both unbaked and baked gabbros carry stable magnetizations. As with metavolcanics, baking has clearly affected the gabbros, since baked gabbros show higher susceptibility and remanence values than the unbaked gabbros (Fig. 3, Table 1). This is most probably due to the fact that the baked gabbros have been heated during the emplacement of the diabase dyke, when some alteration in mineralogy in high temperatures took place (oxidation, growth of magnetic minerals, etc.). The magnetic properties of the gabbros are discussed more briefly in the context of the baked-contact test. Granite samples show somewhat higher Q-values than metavolcanic rock samples, but palaeomagnetic study showed that their magnetization is not stable.

Rock magnetism – magnetic carriers of remanence

Figure 4 shows examples of results from rock magnetic studies.

Thermomagnetic analyses. Representative examples of the Curie point analyses are shown in Figure 4a. These analyses were carried out for one diabase sample from each site, except for sites Haltiavaara (HA1-HA2) and Jokinenä (JO1-JO3), where several samples were analysed. The samples showed variable irreversibilities of heating and cooling curves.

Table 1. *Petrophysical properties of diabase, gabbro and metavolcanic rocks of Salla area*

Rock type	B/N/n	D (kg/m^3)	K (uSI)	J (mA/m)	Q
Diabase	15*/197/780	2981 ± 22	17087 ± 3697	695 ± 340	1.0 ± 0.3
Baked gabbro	1/12*/61	2973 ± 66	39263 ± 19064	2238 ± 1602	1.6 ± 1.4
Unbaked gabbro	2/16*/65	3133 ± 22	9447 ± 7873	420 ± 443	0.8 ± 0.4
Baked metavolcanic rocks	3/23*/77	2792 ± 83	11449 ± 13449	447 ± 580	0.5 ± 0.4
Unbaked metavolcanic rocks	6/28*/130	2936 ± 41	656 ± 80	5 ± 3	0.2 ± 0.2
Granite	1/6*/20	2617 ± 2	262 ± 125	10 ± 3	1.3 ± 1.2

B/N/n is the number of sites/samples/specimens, where *denotes the level used in statistical calculations. Lat, latitude; Long, longitude of the sampling site; D, density; K, susceptibility; J, the intensity of the natural remanent magnetization; Q, the Koenigsberger's ratio calculated for a field of 50 μT. The standard deviations are calculated for each for the mean value.

For example, sample ST64 from HA2 (60 cm-wide diabase apophysis) shows strong irreversibility, indicating mineralogical alteration during heating. In contrast, samples JN1 (site JO3) and JS2 (site JT) are nearly reversible, indicating that these samples are suitable for thermal demagnetization. A few samples show only one ferromagnetic phase with Curie points of 550–580 °C (e.g. sample JS2; Fig. 4a). However, the majority of the samples show two ferromagnetic phases in their thermomagnetic curves: one with Curie temperatures at *c*. 300–340 °C and a second one with Curie temperatures at *c*. 550–580 °C. The lower temperature phase could be associated with pyrrhotite or titanomagnetite, whereas the latter one points to magnetite with varying Ti content. The prominent peak of a lower temperature phase in the sample ST64 indicates the presence of pyrrhotite. The concentration of pyrrhotite varies in samples, which is seen in thermomagnetic curves (Fig. 4a) as differing heights of the lower temperature peak with respect to general behaviour. When the thermomagnetic curves show pyrrhotite there is also the likelihood that the samples contain some pyrite. During heating, pyrite may oxidize to pyrrhotite and magnetite, which can be seen as the irreversible thermomagentic curves.

Scanning electron microscope studies. Scanning electron microscope studies verified the presence of magnetite, with variable Ti content, pyrrhotite and pyrite in the Salla Dyke samples. Examples of these are shown in Fig. 4b (lower image). Often within the same sample some magnetite grains look fairly fresh showing their primary shape (e.g. JS2 in Fig. 4b), whereas other magnetite grains

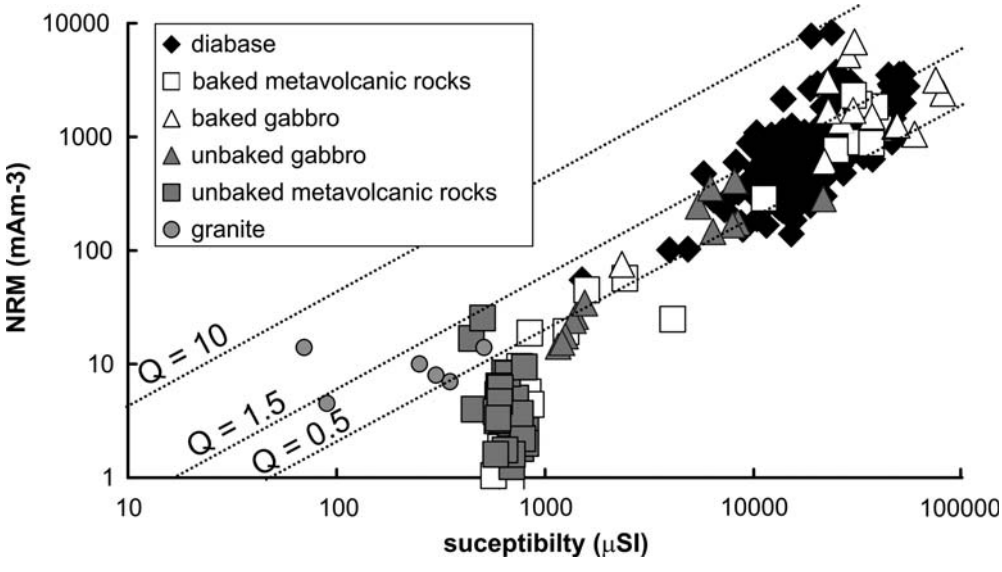


Fig. 3. Intensity of natural remanent magnetization v. magnetic susceptibility with equal Q lines of Salla diabase and host rock samples.

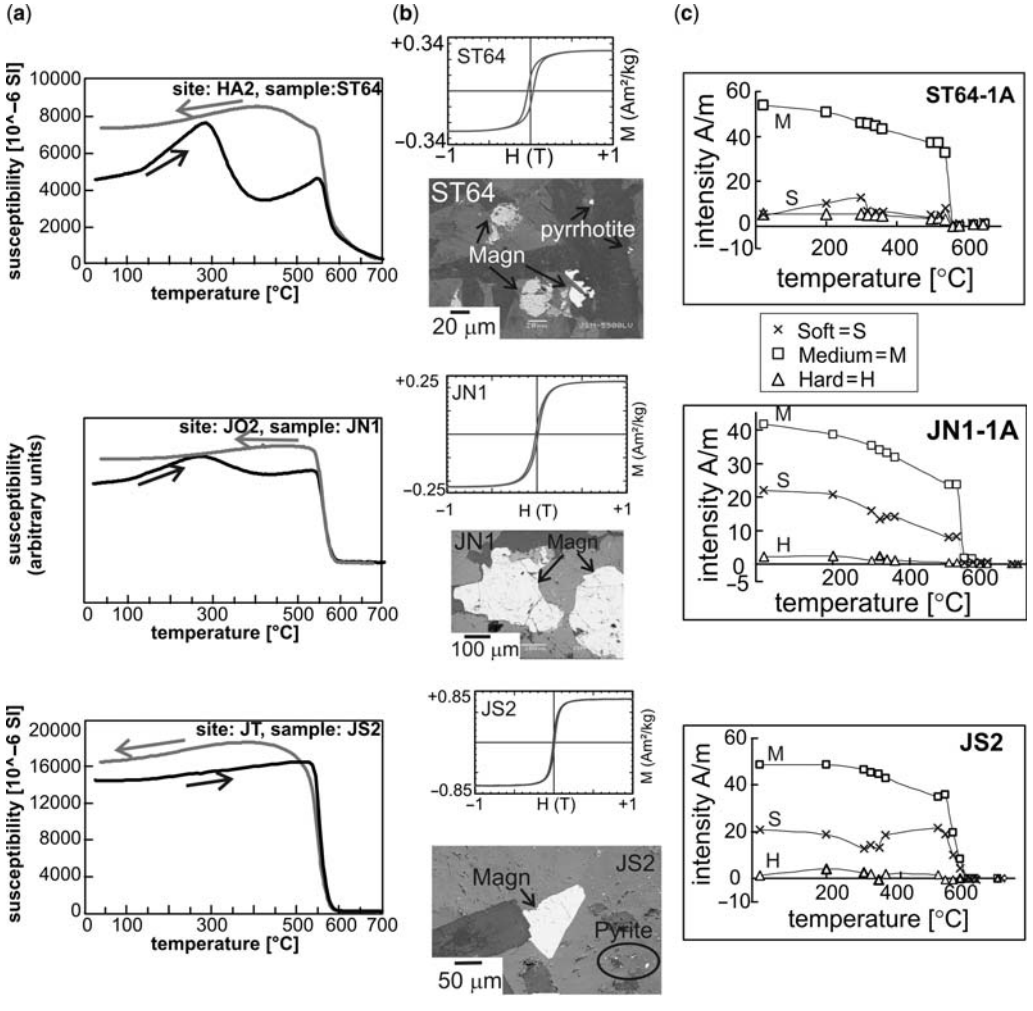


Fig. 4. Rock magnetic properties of the samples from Salla Diabase Dyke. **(a)** thermomagnetic curves (susceptibility v. temperature), **(b)** hysteresis loops (upper image) and Scanning Electron Microscope figures of magnetic minerals (lower image), **(c)** Lowrie test for the samples.

appear to be highly altered (e.g. ST64 and JN1 in Fig. 4b).

Hysteresis properties. Typically magnetite saturates at 0.3 T, titanomagnetites around 0.1–0.2 T and pyrrhotite at 0.5–1.0 T (O'Reilly 1984; Dekkers 1988). The hysteresis loops show that all the diabase specimens saturate in high fields up to 0.6 T (Fig. 4b, upper image), indicating the probable presence of soft and hard carriers of magnetization. Since there are both (titano)magnetite and pyrrhotite in the same samples, we would expect the presence of wasp-waisted hysteresis loops, but this is not observed. Thus we suppose that the amount of pyrrhotite is relatively small compared with the amount of (titano)magnetite. However, the presence of pyrrhotite makes the use of the hysteresis ratios (such as M_r/M_s and H_c/H_{cr}, where M_r is saturation remanence, M_s is saturation magnetization, H_c is coercive force and H_{cr} coercivity of remanence) and the Day plot (Day *et al.* 1977) less usable in terms of defining the magnetic domain sizes.

Lowrie test. In order to study the coercivities of ferromagnetic minerals, we performed the Lowrie test (Lowrie 1990) for the selected diabase samples. Isothermal remanent magnetization (IRM) was given in three orthogonal directions (see Methods above) and the intensity curves of thermal demagnetization of each component were plotted separately (Fig. 4c).

The nine studied specimens from six different sites show consistent results. In all samples the IRM of the components decay to zero below 600 °C, indicating that there is no sufficient amount of hematite in the samples. The intensities of the soft and hard components decay clearly around 300–330 and 520–580 °C, pointing to pyrrhotite and magnetite/titanomagnetite, respectively. The medium fraction dominates the samples. The intensity drops around 520–580 °C, and indicates that magnetite/titanomagnetite is the dominant magnetic mineral. Results thus indicate that medium- to soft-coercivity magnetite/titanomagnetite dominates and pyrrhotite exists to a lesser extent.

Anisotropy of susceptibility

For reliable palaeomagnetic studies, the obtained characteristic remanence direction should reflect accurately the Earth's ambient magnetic field direction at the time of acquisition. It has been suggested that the direction of remanence will not be significantly altered if the ratio between the maximum and minimum suceptibilities (degree of anisotropy) is lower than 5% (Hrouda 1982). Anisotropy of magnetic susceptibility (AMS) can record the flow direction of the magma, but later deformations may change it and cause additional deviations for the remanence directions (Dragoni *et al.* 1997; Elming & Mattsson 2001). AMS was measured on 109 diabase specimens. Measurements were conducted for both non-demagnetized and alternating-field (AF) demagnetized samples, since AF demagnetization does not cause significant changes in AMS (Puranen *et al.* 1992).

For Salla Diabase Dyke the degree of anisotropy is typically below 5%, indicating that the samples have no significant AMS, and the samples are thus suitable for palaeomagnetic study. Both magnetic lineation (L) and foliation (F) values are below 1.04, indicating that the dyke has not been affected by later deformations (Tarling & Hrouda 1993). The AMS data can be used to define the direction of magma flow (Dragoni *et al.* 1997; Elming & Mattsson 2001). When a basic magma flows in the crust, the magnetite grains are aligned with the long axes parallel to the direction of magma flow and the short axes perpendicular to it. In the Salla Dyke, the directions of maximum axis are highly variable, but the majority of the samples show shallow to moderate inclinations with no preferred orientation (e.g. strike of the dyke). The directions of the minimum axes vary greatly. Samples close to the contact show similar directions to the samples taken from the middle of the dyke. The problem with the Salla Diabase samples is that the concentration of pyrrhotite varies (discussed above). Large concentration of pyrrhotite may cause some biased AMS, which will not reflect the original magma flow. Wide scatter in directions of maximum and minimum axes may be due to two factors. First, pyrrhotite may affect the results, obscuring the original magma flow fabric; or second, the magma flow may have been originally turbulent (Hrouda 1982; Tarling & Hrouda 1993).

Palaeomagnetic results of diabase samples

The palaeomagnetic results of the diabase dyke and host rock sites are shown in Table 2. Examples of demagnetization behaviour of the diabase specimens are shown in Figure 5. Palaeomagnetic results of the other host rock sites are presented in the section 'Baked-contact test' below.

The diabase samples carry two remanence components. The first one is typically demagnetized during the first steps of alternating-field (AF) treatment (0–10 mT) and below 340 °C (Fig. 5a–c). It has variable directions and probably represents a viscous remanent magnetization (VRM). The majority of the samples (170 out of 194 samples) also show a harder, characteristic remanence (ChRM) with steep downward direction. The ChRM is isolated below 580 °C, suggesting that remanence resides in magnetite (Fig. 5d). Diabase samples responded similarly for both AF and thermal treatments (Fig. 5). In some specimens, however, AF cleaning was more effective in isolating the ChRM from the viscous component. AF treatment shows that the remanence of diabases is mainly fairly soft (Fig. 5a, c), yet the direction is stable. The finer-grained diabases show somewhat harder behaviour (Fig. 5b). The mean direction of the ChRM is $D = 42.2^\circ$, $I = 73.9^\circ$ ($k = 75.7^\circ$, $\alpha_{95} = 4.8^\circ$; 13 sites; Fig. 7). When using the reference location for central Baltica (Kajaani city, Lat = 64°N and Long = 28°E), the mean direction is $D = 39.8^\circ$ and $I = 73.6^\circ$.

Results of the viscosity test (Dunlop & Buchan, 1977) on two specimens from site Haltiavaara 2 and Jokinenä 3 show that viscosity coefficients S_v of these specimens (sample ST44-3c: 3.0% and sample JM5-2b: 4.9%) are below 5%. In general, values below 5% imply that the remanent magnetization is stable and not due to viscous magnetization (Hrouda 1982; Tarling & Hrouda 1993).

Baked-contact test

Baked-contact tests can be used to verify that the observed remanence in the dyke is thermoremanence with a primary ambient field direction acquired during emplacement (Everitt & Clegg 1962). The test is based on the idea that an igneous intrusion heats the host rock at the contact above the unblocking temperatures of magnetic

Table 2. *Palaeomagnetic results for the Salla diabase dyke and for the gabbro sill*

Site	Site	Lat/Long	B/N/n	D (°)	I (°)	k	α_{95} (°)	Plat (N°)	Plon (E°)	*dp* (°)	*dm* (°)	A_{95} (°)	K	S (°)
					diabase									
Haltiavaara 1	HA1	66.75/28.97	66*/199	34	79	6	9	77	93	16	17	16		
Haltiavaara 2	HA2	66.75/28.97	7*/23	36	63	34	11	60	151	13	16	15		
Helppilä	HE	66.82/28.75	2*/3	48	87			71	43					
Hanhivaara 1	HV1	66.71/29.17	5*/14	52	63	5	37	55	131	46	59	52		
Hanhivaara 2	HV2	66.7/29.2	5*/8	25	69	41	12	71	157	17	21	19		
Jokinenä 1	JO1	66.83/28.58	5*/23	53	70	10	25	63	127	38	44	41		
Jokinenä 2	JO2	66.83/28.58	14*/40	40	80	20	9	75	86	17	18	17		
Jokinenä 3	JO3	66.83/28.5	16*/59	83	70	15	10	51	96	15	17	16		
Jormantie	JT	66.82/28.62	7*/20	32	76	56	8	77	111	14	15	15		
Kulvakkojänkä	KU	66.92/28.18	9*/16	36	69	26	10	68	139	15	18	16		
Pyhäjärvi	PY	66.8/28.8	11*/19	34	72	9	16	72	134	25	29	27		
Pahakuru	PA	66.71/28.88	18*/48	19	88	23	7	71	33	15	15	15		
Possolijärvi	PO	66.7/29.2	5*/12	38	67	37	13	64	142	17	21	19		
		Mean	**13*/170/484**	**42**	**74**	**76**	**5**	**71**	**113**	**8**	**9**	**8**	**27**	**15.6**
				40 (D_r)	74 (I_r)									
					baked gabbro									
Jokinenä 1-3	JO1-3	66.83/28.5	**1/14*/51**	**62**	**74**	**50**	**6**	**63**	**100**	**9**	**10**	**9**	**20**	**34**
				60 (D_r)	74 (I_r)									
					unbaked gabbro									
Jokinenä 1-3	JO3	66.83/28.5	**1/6*/16**	**338**	**57**	**73**	**8**	**59**	**243**	**8**	**12**	**9**	**53**	**12**
60 m to dyke				338 (D_r)	55 (I_r)									
Jokinenä 4	JO4	66.82/28.58	**1/5*/9**	**338**	**61**	**38**	**13**	**63**	**245**	**15**	**19**	**16**	**24**	**17**
2 km to dyke				339 (D_r)	59 (I_r)									

Note: Mean pole (direction) has been calculated as a mean for poles (directions) of all the sites. D_r (I_r) is the declination (inclination) calculated for Baltica's reference location (Lat: 64°N, Long: 28°E; Pesonen *et al.* 2003). Lat and Long are the latitude and longitude of the sampling site. B/N/n denotes number of sites/samples/specimens, where *denotes the statistical level used for mean calculations. D and I are the declination and inclination of the remanent magnetization component. k is Fisher's (1953) precision parameter for the direction. α_{95} is radius of circle of 95% confidence of directions. Plat and Plon are the latitude and longitude for the virtual geomagnetic poles pole. *dp* and *dm* are the semi axes of the oval of 95% confidence of the pole. A95 is the radius of the circle of 95% confidence of the pole. K is Fisher's (1953) precision parameter for the pole. s is the estimated angular standard deviation.

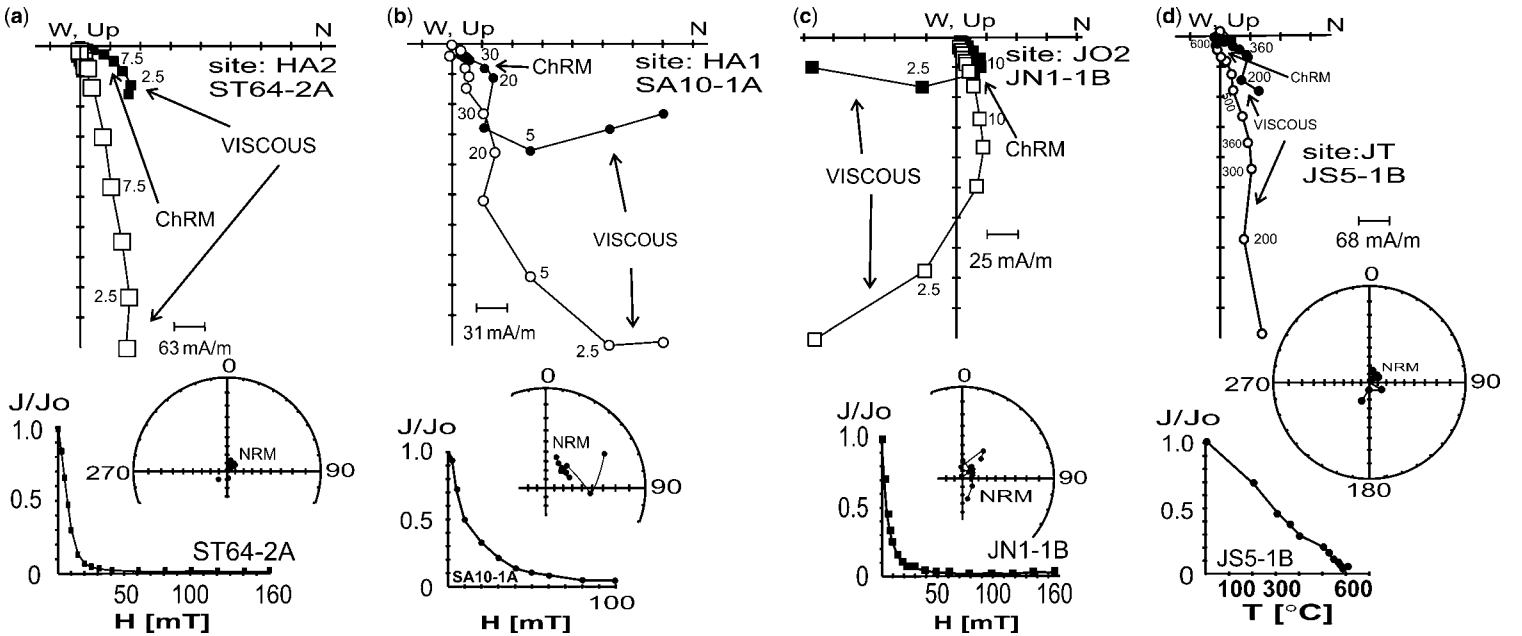


Fig. 5. Examples of alternating field (**a**–**c**) and thermal **(d)** demagnetization behaviour of specimens of the dyke. Upper row is showing orthogonal (Zijderveld) demagnetization diagrams. Open (closed) symbol denotes vertical (horizontal) plane. Middle row is showing stereoplots of observed direction of magnetization. Lower row is showing the relative intensity decay. Numbers in Figure a–c (d) denotes used cleaning af-field (temperature).

minerals. When cooling, the remanence carriers will acquire the same direction as the igneous intrusion (e.g. dyke). Farther away from the intrusion, the host rock retains its pre-intrusion remanence. In between, the samples can hold two remanences that are superimposed: the pre-intrusion remanence and the intrusion remanence (Everitt & Clegg 1962; McCelland-Brown 1981).

The gabbro and the metavolcanic samples close to the Salla Dyke were collected for a baked-contact test at site Jokinenä (JO1-4). For comparison, several Archaean unbaked metavolcanic and granite samples were collected far from the dyke. However, due to unstable behaviour during demagnetization, the metavolcanic samples failed to provide reliable data. Also, the host rocks collected for comparison show weak stability of the remanence and could therefore not be used to verify the primary nature of the dyke.

Only the *c*. 2.2 Ga-old gabbro samples provided magnetically stable results among sampled host rocks. They clearly give a positive baked-contact

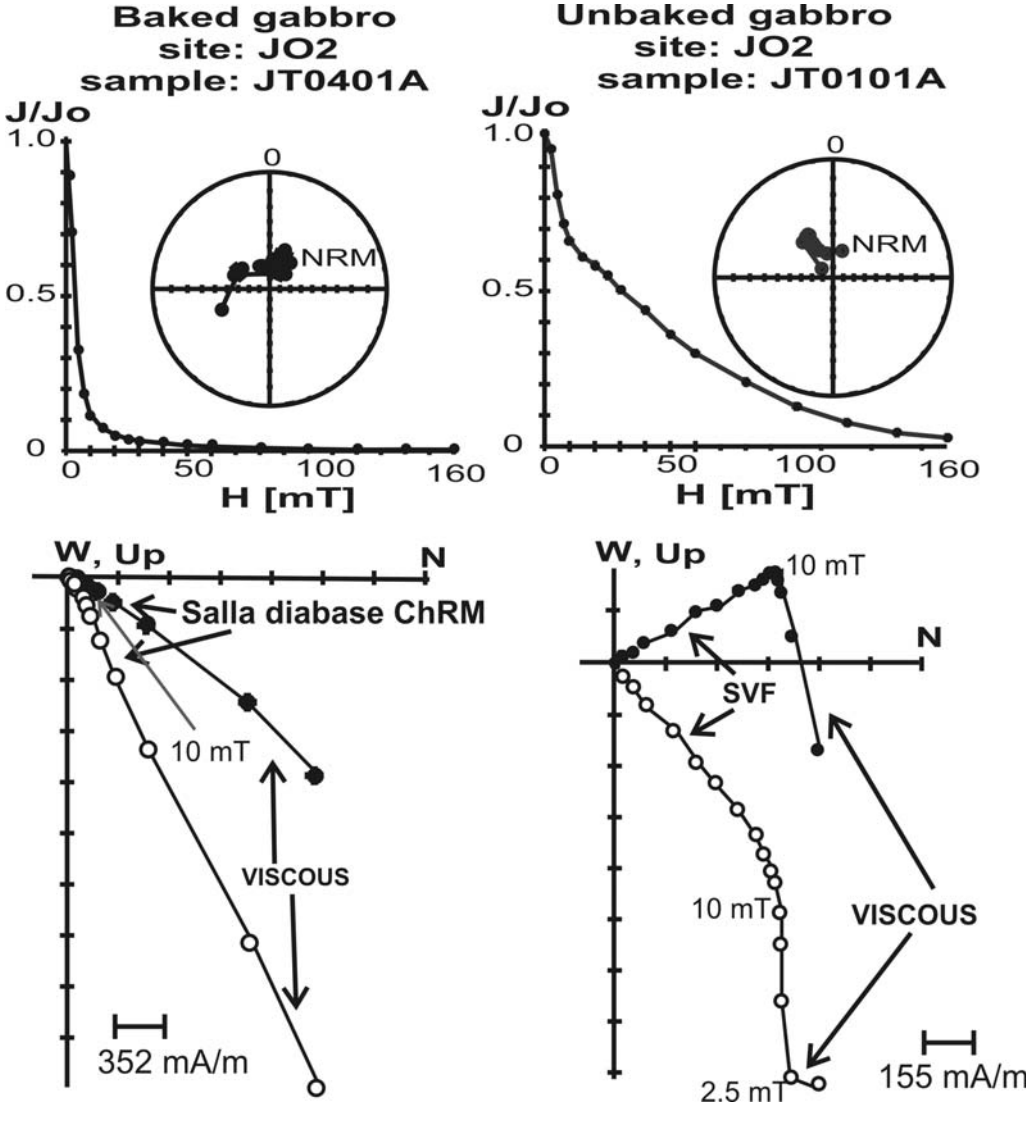


Fig. 6. Examples of positive baked-contact test. Left: baked gabbro (2.2 Ga) taken 30 m from diabase dyke at site Jokinenä. Right: unbaked gabbro taken 60 m from diabase dyke at site Jokinenä. Upper figures show the relative intensity decay and stereographic projections of directional data upon demagnetization. Lower figures show orthogonal (Zijderveld) demagnetization diagrams. Open (closed) symbol denotes vertical (horizontal) plane.

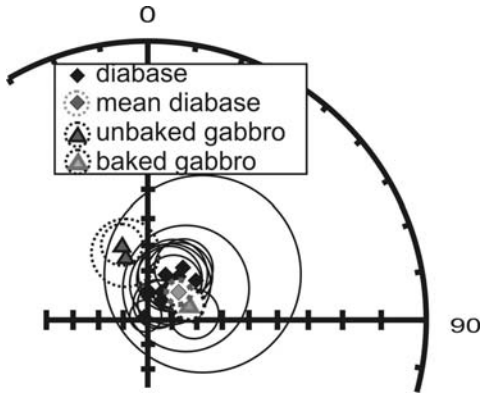


Fig. 7. Palaeomagnetic directions for diabase, baked gabbro and unbaked gabbro samples (Table 2), equal angle projection.

test (Figs 6 & 7). At site JO1-3, where the dyke width is 60–100 m, the remanence direction of the gabbro samples within 30 m of the contact is similar to that of the diabase dyke itself ($D = 62^\circ$, $I = 74^\circ$; Figs 6 & 7; Table 2). The baking effect is also seen in the increased remanence intensity and magnetic susceptibility values in the gabbro samples close to contact, whereas the unbaked gabbro samples show distinctly lower values (Fig. 3, Table 1). The unbaked gabbros, some 60 m away from the dyke (site JO3), show clearly different direction ($D = 338^\circ$, $I = 57^\circ$) from the baked gabbros, and their magnetization is harder (Fig. 6). Similar direction ($D = 338^\circ$, $I = 61^\circ$) was observed also from the gabbros 2 km away from the dyke (Fig. 7). The unbaked direction is similar to the typical Svecofennian *c.* 1.9–1.8 Ga remanence direction, which is found widely throughout Fennoscandia (Mertanen 1995). We interpret this direction to be an overprint caused by the Svecofennian Orogeny. However, the baked-contact test verifies that the Salla Dyke clearly carries a different natural remanent magnetization component than the host rocks.

Discussion

The characteristic, steep, northeastward pointing, remanence direction ($D = 42.2^\circ$, $I = 73.9^\circ$; Fig. 7) for Salla Diabase Dyke was separated from 170 samples (484 specimens). The ChRM yields a virtual geomagnetic pole (VGP) at Plat = 71°N and Plon = 113°E ($A_{95} = 8.1^\circ$).

The 95%-error circle of this ChRM component (Table 2) encloses the direction of the Present Earth's field ($D = 11^\circ$, $I = 76^\circ$) on the sampling site. However, on the basis of the positive baked-contact test to the 2.2 Ga gabbro sill at site Jokinenä, on the fact that the diabase carries clearly two components (viscous and stable ChRM; Fig. 5) and on the viscosity test that indicates stable remanent magnetization, we suggest that the origin of the ChRM component is thermal and represents the primary remanence of the diabase dyke acquired during the cooling of the dyke. Since it has been interpreted that the dyke was emplaced near the surface of the Earth (Väänänen 1965) where the cooling of the magma is relatively rapid, the U–Pb age (1122 ± 7 Ma; Lauerma 1995) of the dyke can be correlated with the timing of the remanence acquisition. Rock magnetic and microscopic studies (Curie point, hysteresis, Lowrie test and SEM studies, Fig. 4) show that magnetite in a wide range of coercivities is the carrier of the primary remanence.

Secular variation

In order to reconstruct the palaeogeographical position of Baltica during the magnetization of Salla Diabase Dyke, a virtual geomagnetic pole (VGP) was calculated for the characteristic remanence component (Table 2). Although the Salla Dyke is a fairly thick, vertical dyke intrusion (width 60–170 m and length 160 km) with apophysic dykelets, it is possible that in the case of only one dyke, that sampling does not average out secular variation (SV) (Buchan & Halls 1990). If the SV is not averaged out, it could cause additional error in the estimation of pole position. The difference between individual virtual geomagnetic poles (VGP) and the rotation axis/palaeomagnetic reference frame can be as much as 15°–20° (McElhinny & McFadden 2000).

In the case of one dyke, the only way to study if SV has been averaged out is to calculate the angular dispersion (between-site dispersion, s) of the site-mean virtual geomagnetic poles (Merrill & McElhinny 1983) at the site palaeolatitude. The s-value was calculated from K (dispersion of poles) and upper and lower limits of (s) were calculated using the methods by Cox (1969). The angular dispersion of Salla was calculated and compared with theoretical dispersion, for example with the model G by Merrill & McElhinny (1983) for the palaeolatitude of the sampling site (60°). We obtained a value of $s = 15.6^\circ + 4.7^\circ/-2.9^\circ$ (Fig. 8) for the dispersion. Even though the upper limit falls on the model G curve, the mean (s) falls well below the SV-model curve of Merrill & McElhinny (1983), suggesting that the secular variation may not have been adequately averaged out. Therefore this might cause an additional error both to the location of the VGP of the Salla Dyke, and as a consequence, to the ancient location of Baltica based on this VGP.

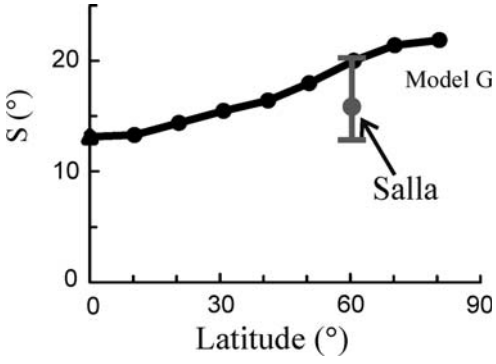


Fig. 8. Calculated between site dispersion of palaeomagnetic pole for Salla Diabase Dyke is plotted on the mean curve (model G) for secular variation, after Merrill & McElhinny (1983).

Implications for pre-Sveconorwegian apparent polar wander path of Baltica

The virtual geomagnetic pole (VGP; 71°N, 113°E) of the characteristic remanence component of the Salla Diabase Dyke is not close to any known Mesoproterozoic poles of Baltica (Fig. 9, Table 3). This VGP refines Baltica's apparent polar wander path (APWP) and creates a loop to the APWP between 1265 and 1042 Ma, just older than the Sveconorwegian Loop. Since we are using a VGP, which may differ from a real palaeopole by 15–20° (McElhinny & McFadden 2000) plus its own A_{95} (8.1°), the suggested loop may be anything between *c*. 50° and *c*. 100° (Salla Dyke VGP is used in Fig. 9 and drawn loop is 70°). After *c*. 1265 Ma the APWP makes a clockwise loop towards the Salla VPG until reaching again the equatorial 1100–1040 Ma poles (Bamble Intrusion and Laanila-Ristijärvi Dyke, Table 3). Before 1265 Ma, the path shows a small hinge.

The part before the Grenville Loop in Laurentia's APWP, called the Logan Loop (poles between 1267–1060 Ma; Robertson & Fahrig 1971; Pesonen 1978; Ernst & Buchan 1993), shows a similar shape to the pre-Sveconorwegian Loop of Baltica (Fig. 9). However, the older 1270–1110 Ma limb of the Logan Loop is poorly constrained (Ernst & Buchan 1993; Donadini 2007). The minimum length of the Logan Loop is 40° (Ernst & Buchan 1993; Donadini 2007), but there are only a few data between the older part of the loop, for example the 1235 Ma pole from Sudbury Dykes, and the younger part, for example the 1112 ± 2.4 Ma pole from Seagull Pluton (Table 3). There is one 1141 ± 2 Ma reliable pole from the Abitibi Dykes (Ernst & Buchan 1993). Similar to Baltica, the APWP of Laurentia also shows the small hinge before 1270 Ma (Fig. 9b). Kalahari-Grunehogna craton's APWP also shows a large loop with length of *c*. 60° during the time interval 1165–1000 Ma (Fig. 9c). Unfortunately there are large gaps in the APWP of Kalahari (Gose *et al.* 2004; Hanson *et al.* 2004). In particular, there are no palaeomagnetic data either between 1270 and 1165 Ma or between the combined Umkondo pole (*c*. 1109 Ma) and younger poles from the Namaqua-Natal metamorphic belt (Okiep copper district, Ntimbankula, Namaqua mean pole and Port Edward, Table 3). New palaeomagnetic data from well-dated rocks between 1270–1112 Ma are required from all three continents to test the similarity between Logan Loop of Laurentia, pre-Sveconorwegian Loop of Baltica and the 1.27–1.0 Ga segment of Kalahari-Grunehogna's APWP. All these show large APWP loops, which

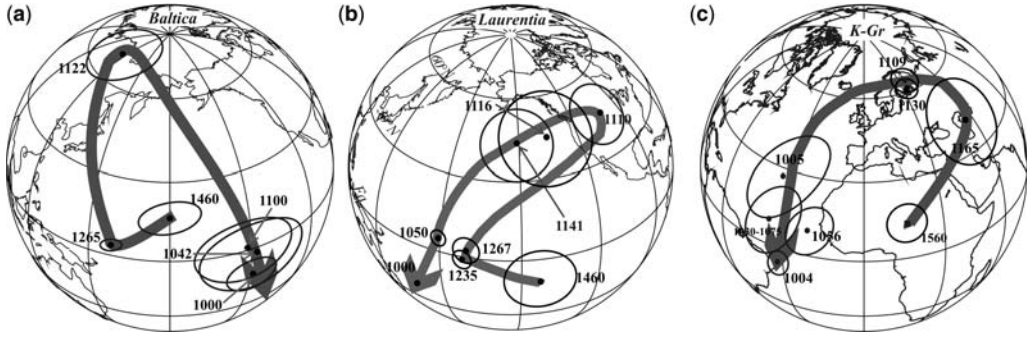


Fig. 9. **(a)** Selected Pre-Sveconorwegian poles for Baltica in present-day Baltican coordinates. Numbers are ages in Ma. **(b)** Logan Loop of Laurentia (Ernst & Buchan 1993; Donadini 2007) based on selected poles for Laurentia in present-day Laurentian coordinates, **(c)** Palaeomagnetic poles for the Kalahari-Grunehogna craton in the 1165–1000 Ma interval in present-day African coordinates. Grunehogna reconstructed to Africa by Euler parameter: Lat: −9.7°, Long: 328.8°, angle 56.3° (Powell & Li 1994). Used poles are listed in Table 3.

Table 3. *Selected Late Mesoproterozoic palaeomagnetic data from Baltica, Laurentia and Kalahari–Grunehogna cratons*

Pole	Age (Ma)	Plat	Plon	*dp*	*dm*	Reference
Baltica						
Mean Baltica	1512	39	171	16	16	Pesonen *et al.* 2003
Mean Baltica	1460	15	181	11	11	Salminen & Pesonen 2007
Mean Baltica	1265	4	158	4	4	Pesonen *et al.* 2003
Salla Diabase VGP	1122 ± 7	71	113	8	8	this work
Mean Baltica	1100	1	208	16	16	Pesonen *et al.* 2003
Laanila-Ristijarvi	1042 ± 50	−2.1	212	13	21	Mertanen *et al.* 1996
Protogine zone Dolerites	1000	−12	211	7	7	Pisarevsky & Bylund 1998
Laurentia						
Mean Laurentia	1460	−7	217	13	13	Salminen & Pesonen 2007
Mean MacKenzie Dykes	1267 + 7/−3	4	190	5	5	Buchan & Halls 1990; Fahrig *et al.* 1981; Le-Cheminant & Heaman 1989
Sudbury Dykes	1235 + 7/−3	−3	192	3	3	Schwarz & Buchan 1982; Dudas *et al.* 1994
Abitibi Dykes (N)	1141 ± 2	43	209	13	13	Ernst & Buchan 1993; Krogh *et al.* 1987
Seagull Pluton	1112 ± 1.4	42	233	2	2	Borradaile & Middleton 2006; Davis & Paces 1990
Arizona Dykes (R)	1110	46	257	12	12	Donadini 2007
Logan Sills	1109 + 4/−2	49	220	4	4	Pesonen 1978; Davis & Green 1997
Nipigon Inspiration Sills	1109	45	218	2	2	Borradaile & Middleton 2006
Mean Laurentia	1116	46	227	13	13	this work
Mean Laurentia	1099	36	188	–	–	Pesonen *et al.* 2003
Nonesuch Shale + Freda sandstone	1050	6	179	3	3	Pesonen *et al.* 2003
Mean Laurentia	1000	−20	167			Pesonen *et al.* 2003
Kalahari-Grunehogna						
Van Dyk Mine Dike	1560–1610	12	14	6	8	Jones & McElhinny 1966
Premier Kimberlites	1165 ± 10	41	55	16	16	Powell *et al.* 2001
Ritscherflya Supergroup	1130 ± 10	−8	232	4	4	Powell & Li 1994;
rotated to Kalahari		−61 (rot)	29 (rot)			Powell *et al.* 2001
Borgmassivet Intrusion	1130 ± 10	8 (−8)	55 (235)	3	3	Jones *et al.* 2003
rotated to Kalahari		62 (rot)	35 (rot)			
Mean Grunehogna	1130 ± 10	−8	234	4	4	Jones *et al.* 2003
rotated to Kalahari		61 (rot)	32 (rot)			
Umkondo combined	1109 + 2/−1	64	36	2	2	Hanson *et al.* 2004
Okiep copper district	1056 ± 20	8.8	339.5	7.6	7.6	Muller *et al.* 1978
Namaqua mean pole	1030–1075	9.4	324	13	13	Onstott *et al.* 1986
Ntimbankula pluton	1005	28	325	15	15	Mare & Thomas 1998
Port Edward pluton	1004 ± 5	−7.4	327.8	4.2	4.2	Gose *et al.* 2004

Plat and Plon are latitude and longitude of the palaeomagnetic pole. *dp* and *dm* are the semi-axes of the oval of 95% confidence of the pole. 'Rot' means that pole has been rotated.

appear to reach their apexes at different ages (Fig. 9). It is evident that Baltica, Laurentia and Kalahari experienced fast drift rates during these time intervals, but we need more coeval well-dated palaeomagnetic poles to test similarities between these APWPs.

1.12 Ga palaeoreconstruction of continents

Using the new 1.12 Ga virtual geomagnetic pole (VGP) for Baltica, we test the model of continuation of the proposed long-lasted unity of Baltica and Laurentia during the Mesoproterozoic (Pesonen *et al.* 2003; Salminen & Pesonen 2007 and refs therein). The 1122 Ma Salla VGP takes Baltica to a high palaeolatitude of *c*. 60° (Fig. 10a), but since we are using a VGP, Baltica may also occupy a palaeolatitude of 30° (see previous section). Laurentia's position is based on a 1116 Ma mean pole (Table 3), and it also occupies high palaeolatitudes. Palaeoreconstruction in Figure 10a is made using the individual *c*. 1.12 Ga poles to reconstruct each continent to its correct palaeolatitudinal position and evaluate the relationships between continents based on the 'closest approach' technique, which takes into account the maximum probable error limits of the poles. We use the 1.12 Ga Salla VGP for Baltica, where the error bar includes errors from both secular variation (15–20°; McElhinny & McFadden 2000) and A_{95} of Salla VGP (8.1°); the mean 1.12 Ga pole for Laurentia; the 1.14 Ga Mount Isa Lake View Dolerite pole for Australia; and the 1.13 Ga mean pole for Kalahari-Grunehogna (Table 4). Since the poles for Baltica and Laurentia do not have exactly the same age (Table 4) it is permissible that their ovals of confidence are just overlapping (Fig. 10a). However, if we calculate Baltica's Euler rotation in relation to Laurentia, we get Euler parameters of Lat: 53°, Long: 0.5° and angle +58°.

There are not exactly any coeval palaeomagnetic data for other continents. Closest in ages are Australia, Kalahari and Grunehogna (Table 4). Australia is reconstructed using the pole from Mount Isa Lake View Dolerite, which places Australia also at high palaeolatitudes, as in Pesonen *et al.* (2003). Other plates (Kalahari–Grunehogna) seem to occupy equatorial palaeolatitudes (both polarity options are shown) at 1.12 Ga.

Proposed long-lasting connection between Baltica and Laurentia during the Mesoproterozoic

Recent 1.46 Ga palaeomagnetic data of Salminen & Pesonen (2007) support the idea of Baltica–Laurentia unity between 1.76 and 1.27 Ga. We use the new VGP of Salla to test this proposed long-lasting unity between Baltica and Laurentia at 1.76–1.12 Ga. Moreover, this connection between Baltica and Laurentia also formed the core of the Mesoproterozoic supercontinent Hudsonland (also

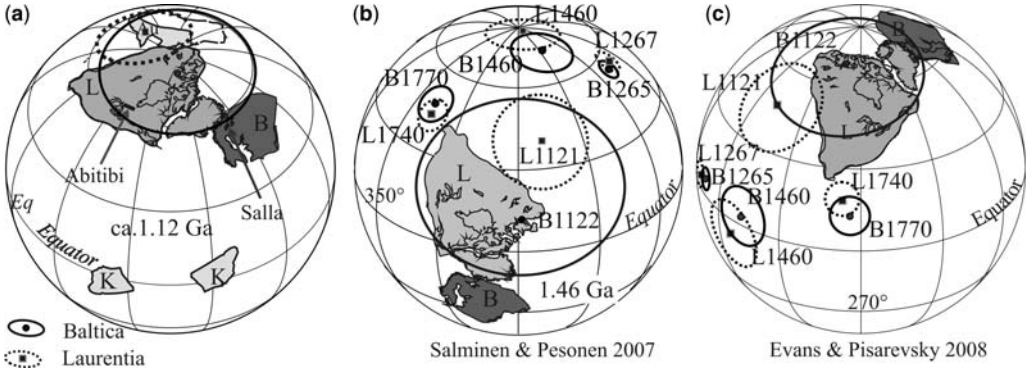


Fig. 10. Palaeoreconstructions of Mesoproterozoic connection between Baltica and Laurentia. Numbers are ages of used poles in million years. B (L) stands for Baltica (Laurentia). **(a)** Reconstruction at 1.12 Ga using the 'closest approach' technique where the continents are shown at their palaeolatitudes. Abitibi and Salla dykes are drawn in Laurentia and Baltica, respectively. The error bar of the Salla pole (B1122) includes the error caused by secular variation. **(b)** Reconstruction of Baltica and Laurentia using the 'closest approach' technique at 1.46 Ga, based on mean poles for Baltica and Laurentia (Euler pole for Baltica is Lat: 0°, Long: 102°, angle 79° and for Laurentia is Lat: 0, Long: 312, angle: −93.5). Also shown 1.76, 1.46, 1.27 and 1.12 Ga poles for Baltica and Laurentia by using this continental configuration (after Salminen & Pesonen 2007). **(c)** Reconstruction according to Evans & Pisarevsky (2008) where Baltica and its poles have been rotated to Laurentia's reference frame using the Euler parameters: Lat: 47.5°, Long: 1.5°, angle: +49°. Also shown 1.76, 1.46, 1.27 and 1.12 Ga poles for Baltica and Laurentia according to this continental configuration. Used data are listed in Table 4.

Table 4. *Palaeomagnetic poles used for reconstructions*

Craton	Formation	Plat (°)	Plong (°)	A_{95} (°)	Age (Ma)	Original references
			c. 1.12 Ga			
B	Salla Diabase Dyke VGP	71	113	8	1122 ± 7	this work
L	Mean Laurentia	46	227	17	1116	this work, see Table 3
Au	Mount Isa Lakeview Dolerite	−10	131	17	1140	Tanaka & Idnurm 1994; unpublished age cited in Wingate & Evans 2003
K–Gr	Mean Kalahari–Grunehogna	61	32	4	1130 ± 10	Jones *et al.* 2003
			c. 1.27 Ga			
B	Mean Baltica	4	158	4	1265	Pesonen *et al.* 2003
L	MacKenzie dykes	4	19	5	1267 + 7/−3	Buchan & Halls 1990, LeCheminant & Heaman 1989
			c. 1.46 Ga			
B	Mean Baltica	15	181	11	*c.* 1460	Salminen & Pesonen 2007
L	Mean Laurentia	−7	217	13	1460	Salminen & Pesonen 2007
			c. 1.77 Ga			
B	Ropruchey–Shoksha	42	221	7	*c.* 1770	Pisarevsky & Sokolov 2001
L	Cleaver Dykes	19	263	6	1740 + 5/−4	Irving *et al.* 2004

B, Baltica; L, Laurentia; Au, Australia; K, Kalahari; Gr, Grunehogna. Plat and Plong are latitude and longitude of the palaeomagnetic pole. A_{95} is the 95% confidence circle of the pole.

known as Columbia or Nuna; Meert 2002; Rogers & Santosh 2002; Pesonen *et al.* 2003; Zhao *et al.* 2003, 2004 and refs therein). Salminen & Pesonen (2007) propose a palaeoconfiguration at 1.46 Ga where northern Baltica is placed against northeastern Laurentia (Fig. 10b). This configuration is made using the 'closest approach' technique, where continents have been rotated to their correct palaeolatitudes. Calculated Euler parameters in relation to the continents' present positions are for Baltica, Lat: 0°, Long: 102°, angle 79°; and for Laurentia, Lat: 0°, Long: 312°, angle: −93.5°. Pole pairs of Baltica and Laurentia at *c.* 1.76 and 1.27 Ga allow the presented configuration (Fig. 10b, Table 4). Within error limits of the used poles, *c.* 1.76 Ga-old geologically similar Trans Scandinavian Igneous Belt (in Baltica) and Yavapai/Ketilidian (in Laurentia) belts become contiguous in this continental reconstruction. Moreover, there is geological and geophysical evidence, including the proposed common magma source for coeval dolerites in north and central Sweden and in Finland, and basalts and dolerites in NE Greenland at 1.27 Ga, supporting such a continental configuration (Elming & Mattsson 2001; Karlström *et al.* 2001).

The 1.12 Ga poles (VGP of Salla Dyke for Baltica and mean 1116 Ma palaeomagnetic pole for Laurentia, Table 3) are not overlapping, but their error ovals (for Salla Dyke VGP 23–28° and for Laurentia 13°; see previous section and Table 3) overlap in this continental reconstruction, which would imply that they allow the configuration. Thus there is new palaeomagnetic evidence for a long-lived (1.76–1.12 Ga) Baltica–Laurentia connection, although not conclusive since VGP is used for Baltica.

This particular configuration is also supported by other studies. Some researchers proposing a long-lived Baltica–Laurentia connection use somewhat similar configuration of Baltica and Laurentia during the Mesoproterozoic (Gower *et al.* 1990). For example, Evans & Pisarevsky (2008) rotate Baltica using Euler parameters: Lat: 47.5°, Long: 1.5°, angle: +49° in relation to Laurentia (Fig. 10c). This is a different approach than in Figure 10b, since Baltica and the poles of Baltica have been rotated to a Laurentian reference frame keeping Laurentia in its present position (based on a hypothetical configuration). Hence, this approach does not show the palaeolatitudes of the continents, although these could easily be computed and illustrated for the joined assembly. The connection between 1.76 and 1.12 Ga has been tested using this configuration (Fig. 10c). If similarly aged poles fall in identical locations following rotation,

then the reconstruction is supported by the palaeomagnetic data. Since there is only slight difference between this and the configuration of Salminen & Pesonen (2007), the results are similar, allowing the Baltica–Laurentia connection between 1.76 and 1.12 Ga. However, we need to take into account the allowances concerning the Salla VGP.

The proposed configuration appears to be close to the standard Rodinia configuration (Pisarevsky *et al.* 2003 and refs therein). This raises a question of whether the Baltica–Laurentia entity lasted until Rodinia. Since the pole from the Salla Dyke is only a VGP, further study of this question requires new, accurately-dated and well-defined palaeomagnetic data.

Conclusions

The well-dated 1.12 Ga Salla Diabase Dyke yields a stable characteristic remanence component. The direction of remanence is close to the known direction of the present Earth's field at the sampling site, but a positive baked-contact test proves its primary nature. Rock magnetic studies indicate that magnetite is the remanence carrier. In spite of 13 different sites along the wide and long dyke for palaeomagnetic data, the results are obtained only from one dyke. Hence it is possible that the secular variation has not been sufficiently sampled out and the pole represents only a virtual geomagnetic pole (VGP).

This VGP is not close to any previously known Proterozoic poles of Baltica. Therefore it refines the pre-Sveconorwegian part of the apparent polar wander path (APWP) of Baltica. It adds a clockwise loop to the Baltica's APWP between 1265 and 1040 Ma, just before the Sveconorwegian Loop. Since the pole is a VGP, which may differ from a real palaeopole by 15–20° (McElhinny & McFadden 2000) plus its own A_{95} (8.1°), the suggested loop may be anything between *c.* 50° and *c.* 100°. A similarly aged Logan Loop (Robertson & Fahrig 1971; Ernst & Buchan 1993; Donadini 2007) of the Laurentian APWP, and the 1165–1000 Ma APWP of Kalahari–Grunehogna, also express long lengths, which may imply a common drift history for these continents. However, new palaeomagnetic data from well-dated rocks between 1270 and 1112 Ma are required from all of these continents to test the similarity between these loops of APWP. More evidence of the common history is the fact that the similarity between Baltica's and Laurentia's APWP already starts at 1.5 Ga. Both APWPs form a hinge after 1.5 Ga when heading to the 1.27 Ga part of the APWP.

The VGP for Salla Diabase Dyke takes Baltica to a high palaeolatitude of *c.* 60° at 1.12 Ga, but since the VGP may differ from a real palaeomagnetic pole (McElhinny & McFadden 2000) the position of Baltica may also be as low as *c.* 30°. At that time, Laurentia and Australia were located at high palaeolatitudes. Unfortunately, besides these examples at *c.* 1.12 Ga, there are palaeomagnetic data only for Kalahari-Grunehogna, which is located in equatorial palaeolatitudes.

The extension of a proposed long-lived Mesoproterozoic connection between Baltica and Laurentia (Gower *et al.* 1990; Åhäll & Connelly 1998; Karlström *et al.* 2001; Pesonen *et al.* 2003 and refs therein) is permissible according to the new 1.12 Ga palaeomagentic data for Salla Diabase Dyke, and supports a long-lived (1.76–1.12 Ga) Baltica–Laurentia connection. However, when taking into account that the ages of the used poles for Baltica and Laurentia are not exactly coeval, and the data for Baltica come from only one dyke, where secular variation has not been adequately sampled, this interpretation needs further verification.

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